    - Measures the long-term expected reward for being in a specific state and following a particular policy. The value function helps the agent evaluate the desirability of states.
5. **Q-Value (Q)**:
    - Extends the value function by incorporating actions. It represents the expected reward for taking a specific action in a specific state and then following the policy. This is central to algorithms like Q-Learning.

Mathematical Framework of RL

Reinforcement Learning is often formulated as a Markov Decision Process (MDP), defined by:

- **States (S)**: A set of all possible configurations of the environment.
- **Actions (A)**: A set of all possible actions the agent can take.
- **Transition Probability (P)**: The probability of moving to a new state given the current state and action.
- **Reward Function (R)**: The immediate reward received after transitioning to a new state.
- **Discount Factor (γ)**: A factor between 0 and 1 that balances immediate rewards and future rewards.

The agent's goal in an MDP is to maximize the **cumulative discounted reward**: Gt=Rt+1+γRt+2+γ2Rt+3+ ⋯ =∑k=0∞γkRt+k+1G_t = R_{t+1} + γR_{t+2} + γ^2R_{t+3} + \dots = \sum_{k=0}^∞ γ^kR_{t+k+1}

Types of Reinforcement Learning Algorithms

1. **Value-Based Methods**:
    - Focus on estimating value functions, such as Q-Learning and SARSA.